Fuel System

Category: **Automotive** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fuel supply System & Urea Handling (Diesel)

What it is used for

An automotive fuel supply system stores, filters, and delivers fuel to the engine at the pressure and flow required for operation. Diesel systems may also include a separate urea-based diesel exhaust fluid (DEF/AdBlue) system for emissions control; correct identification, safe handling, and leak checks help prevent fire, contamination, and system damage.

Main parts

- Fuel tank and filler assembly
- Fuel lines, hoses, and connectors
- Fuel filter and water separator where fitted
- Fuel pump and pressure-control components
- Injectors or fuel rail depending on engine design
- Diesel exhaust fluid (DEF/AdBlue) tank, pump, lines, and dosing injector where fitted

Before you start

1. Only trained automotive personnel should inspect or service fuel and diesel exhaust fluid systems.
2. Identify the vehicle and engine type, fuel specification, pressure hazards, and manufacturer's service procedure.
3. Park the vehicle in a ventilated workshop, apply the parking brake, and allow hot components to cool as required.
4. Inspect the fuel tank, lines, connectors, filters, pump, and DEF components for visible leaks, damage, or corrosion without touching suspected high-pressure leaks.
5. Confirm that the correct fuel and DEF specification are available and that containers are clean and clearly labelled.
6. Review the required pressure-relief, electrical-isolation, and spill-response procedures before opening any system.

How to use

1. Identify the system layout and locate the fuel supply components and, where fitted, the separate DEF tank and dosing system.
2. Switch off the engine and follow the manufacturer's procedure to isolate electrical power and relieve fuel pressure before servicing.
3. Place suitable drip trays beneath the work area and keep approved absorbent material available for small spills.

4. Inspect low-pressure fuel hoses, clamps, connectors, filters, and tank fittings for damage or leakage.
5. For high-pressure diesel injection systems, follow the exact manufacturer procedure; never loosen a high-pressure line while the engine is running or pressure remains present.
6. Replace fuel filters using the specified part and cleanliness procedure, preventing dirt or water from entering the fuel system.
7. Prime the fuel system using the approved method and check for leaks before starting the engine.
8. Where fitted, inspect DEF tank, filler cap, pump, lines, heater, and dosing components for leaks, crystallisation, or damage.
9. Fill the DEF tank only with the specified diesel exhaust fluid from a clean, dedicated container; never add DEF to the diesel fuel tank or diesel fuel to the DEF tank.
10. Follow the manufacturer's procedure for DEF service, priming, or dosing-system checks; do not mix DEF with other fluids or add unapproved additives.
11. Start the engine only after all serviced components are correctly reassembled and tools have been removed from the engine bay.
12. Observe system operation and warning indicators, and check for leaks without placing hands near high-pressure fuel components.
13. If a leak, warning, abnormal pressure, or engine fault appears, stop the engine when safe and investigate using the approved diagnostic procedure.
14. Clean small spills using the appropriate material, collect contaminated waste, and dispose of it according to workshop and environmental requirements.
15. Record parts replaced, fluids used, test results, and any faults requiring further repair.

⚠ Safety

- Diesel fuel is combustible; keep flames, sparks, smoking, and other ignition sources away from fuel handling areas.
- High-pressure diesel fuel can penetrate skin and cause serious injury; never search for leaks with bare hands and seek immediate medical attention after suspected injection injury.
- Relieve system pressure and follow the manufacturer's isolation procedure before disconnecting fuel lines.
- Wear suitable gloves and eye protection and avoid skin contact or inhalation of fuel and DEF mist.
- Never mix diesel exhaust fluid with fuel, coolant, oil, or other chemicals; DEF contamination can damage emissions equipment.
- Clean spills promptly, prevent fluids from reaching drains, and keep absorbents and appropriate fire equipment available.
- Do not work beneath a vehicle supported only by a jack; use approved lifting supports and follow workshop safety rules.

Care & storage

- Inspect fuel hoses, connectors, filters, tank mounts, and seals at the recommended service intervals.

- Replace fuel filters and water separators according to the vehicle manufacturer's schedule.
- Keep fuel and DEF containers sealed, labelled, and protected from dirt and contamination.
- Clean DEF crystals using the manufacturer's approved method and prevent contamination of connectors and dosing parts.
- Arrange qualified diagnosis if pressure, flow, emissions warnings, or recurring leaks remain abnormal.

Common mistakes to avoid

- Opening a high-pressure fuel line before following the pressure-relief procedure.
- Allowing dirt or water into the fuel system during filter replacement.
- Filling the DEF tank with diesel or adding DEF to the fuel tank.
- Ignoring leaks, warning lights, or contamination and returning the vehicle to service without checks.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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